MacHuna v1.17.0 — User Manual

Broadcast Media Format Converter

DNS Vision Limited — For Vision Mixers, TDs, and Support Engineers

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1. Overview

MacHuna is a macOS application for translating broadcast media assets between formats. It converts video clips, TGA sequences, and still images to Grass Valley Kahuna `.SWS` format, reads and writes Grass Valley K-Frame `.EIF` native clips, and extracts `.SWS` files back to standard formats for use on K-Frame and Sony MVS desks.

It runs natively on macOS.

1.1 What MacHuna Does

Direction	From	To
Kahuna SWS	MOV, MP4, MXF, MKV, AVI, TGA sequences, PNG, BMP, JPG	`.SWS` (Grass Valley Kahuna)
K-Frame EIF	MOV, MP4, MXF, TGA sequences, `.SWS`	`.EIF` (Grass Valley K-Frame), with its `.eaf` audio file when the source has sound

Direction	From	To
K-Frame EIF → SWS	`.EIF`	`.SWS` (Grass Valley Kahuna), with the clip's audio
K-Frame EIF → TGA	`.EIF`	32-bit RGBA TGA sequence
K-Frame TGA	`.SWS`, `.EIF`, video clips	32-bit RGBA TGA sequence
Sony TGA	`.SWS`, `.EIF`, video clips, TGA sequences	32-bit RGBA TGA sequence (Sony MVS naming)
TGA Sequence	TGA sequences, video clips	32-bit RGBA TGA sequence (i□p standards conversion)
QuickTime MOV	`.SWS`, `.EIF`, TGA sequences	ProRes 4444 `.mov` with alpha, and audio where the source has it

1.2 Supported Standards

All nine standards below have been confirmed against K-Watch reference files and verified on a live Grass Valley Kahuna mainframe.

Standard	fps	Region
1080i/50	25	UK / Europe
1080i/59.94	29.97	USA / Japan
1080i/60	30	—
1080p/25	25	UK / Europe
1080p/50	50	UK / Europe
1080p/59.94	59.94	USA / Japan
1080p/60	60	—

1.3 Highlights

- Runs natively on macOS
- Accepts MOV, MP4, MXF, MKV and AVI clips, TGA sequences, and PNG, BMP and JPG stills
- Converts between Kahuna SWS, K-Frame EIF, and Sony MVS formats in one app
- Reads and writes K-Frame `.EIF` native clips, with their `.eaf` audio. Confirmed on a live K-Frame in October 2026, at 50p and 50i
- K-Frame TGA sequences confirmed on a live K-Frame, from video clips, `.EIF` and `.SWS`
- Converts SWS, EIF and TGA sequences back to an editable ProRes 4444 QuickTime, with key and audio
- Built-in Video Player for checking `.SWS` and `.EIF` files without a Kahuna or K-Frame desk

> **NOTE** MacHuna converts and extracts files on your Mac. It does not upload to a mainframe over the network and it does not synchronise projects — getting the finished files onto the desk is done however you do it now.

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2. Installation

MacHuna is distributed as a self-contained `.app` bundle. ffmpeg is bundled inside — no separate installation required.

Download it from dnsvision.tv/machuna. That page always carries the current release, the release notes and this manual.

2.1 First Launch

MacHuna is not signed with an Apple Developer certificate, so the first time you open it macOS says it cannot verify the developer and refuses. This is expected. Three steps, once per machine:

1. Move **MacHuna.app** to your **Applications** folder and double-click it. macOS shows a warning and will not open it. Click **Done**
2. Open **System Settings ▸ Privacy & Security** and scroll down. There is a line about MacHuna being blocked, with an **Open Anyway** button. Click it and confirm with Touch ID or your password
3. MacHuna opens, and opens normally from then on

> **NOTE** Older instructions said to right-click the app and choose **Open**. That no longer works on current versions of macOS, which removed the shortcut: you have to go through System Settings as above.

Checking which version you have. MacHuna's title bar shows it, and **Help ▸ Check for Updates…** tells you whether it is the current release. In Finder, select `MacHuna.app` and press ⌘I: the version is shown in the Get Info panel, and the title bar shows it once running. Before v1.6.20 the bundle reported its version as 0.0.0, so if Get Info says 0.0.0 you are running v1.6.19 or earlier. This matters if you keep more than one copy of the app.

2.2 Settings

MacHuna saves all settings automatically on quit. Settings are stored at:

```
~/.kwatch_settings.json
```

To reset to defaults, quit MacHuna and delete this file.

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3. Main Window

The MacHuna window has three rows:

Row	Purpose
Destination Folder	Where converted files are written
Convert	Add files, choose output format, set what each one is called, convert
Log	Conversion progress and errors, five lines

The Convert row adapts automatically based on what files you open. MacHuna detects the input type and shows only the controls that are relevant.

The **item list** in the middle of the Convert row takes whatever height the window has spare, so making the window taller shows more rows at once. The log stays at five lines: the rows themselves now carry the outcome of a conversion, and the full record is written to the log file in your Destination Folder.

4. Converting Files

The workflow is the same regardless of what you are converting or what you are converting it to:

1. Set your **Destination Folder**
2. Click **Open Files…** and select a folder. Once something is loaded the button reads **Add Files…**, since you are usually adding to the list rather than starting again
3. MacHuna scans the folder, detects the input type, and populates the item list
4. Choose your **Output** format from the dropdown
5. Set any options that appear (standard, field order, etc.)
6. **Give each row a number or name**, or tick **Number sequentially** — see Section 4.4
7. Click **Convert**

A conversion log is written to the Destination Folder when the batch completes.

4.1 Input Type Detection

MacHuna reads the contents of the folder you open and determines the input type automatically:

Files found in folder	Detected as	Available outputs
`.SWS` files only	SWS source	Kahuna SWS, K-Frame TGA, K-Frame EIF, Sony TGA
`.EIF` files only	EIF source	Kahuna SWS, K-Frame TGA, Sony TGA
Mix of `.EIF` and `.SWS` files	EIF + SWS source	Kahuna SWS, K-Frame TGA, Sony TGA

Files found in folder	Detected as	Available outputs
Video files only (MOV, MP4, MXF…)	Video source	Kahuna SWS, K-Frame TGA, K-Frame EIF, Sony TGA, TGA Sequence
TGA sequences and/or stills	TGA / stills source	Kahuna SWS, K-Frame EIF, Sony TGA, TGA Sequence — see note below
Mix of SWS and other files	Error — shown in summary	—

> **NOTE** If you see a "mixed input" error, the folder contains both `.SWS` files and other file types. Move them into separate folders and convert each folder independently.

> **NOTE — stills go to Kahuna SWS only.** A still image is a single frame, not a clip, so MacHuna does not convert stills to K-Frame EIF, Sony TGA or TGA Sequence output. Those outputs need a clip or a TGA sequence. If a still is selected with one of them, MacHuna names the file and stops rather than converting it. Deselect the still, or choose Kahuna SWS. A folder holding both stills and TGA sequences is fine: select the sequence and leave the stills unticked.

4.2 The File List

The file list shows one entry per item to be converted:

- **TGA sequences** are collapsed to a single line: `TNTS201 (30 frames)` — you do not need to select individual frames
- **Video files** appear by filename
- **Stills** appear by filename
- **EIF files** appear by filename

Click entries to select or deselect them. By default, all files are selected.

Building a batch from more than one folder (v1.6.15). Once you have a selection, opening another folder gives you two buttons:

- **Select** — replaces the list with what you have chosen here (the original behaviour)
- **Add to List** — appends it to what is already selected

So you can take two files from one folder, one from another, and convert all three in a single batch. Items already in the list are not added twice, and the summary line next to **Open Files…** reports the whole selection (e.g. `2 folders: 3 video files`).

One batch has to be one kind of job, so you can only add items of a compatible type: media being encoded **to** SWS, or SWS/EIF files being extracted **back out**. Trying to mix the two is refused with an explanation — use **Select** to start a fresh list instead. Adding EIF files to SWS files (or the reverse) is fine.

Cancelling the folder browser leaves your current selection untouched.

4.3 The Item List

Once files are selected, MacHuna shows **one row per item**, and that row tells you what the item is about to be called:

```
BAY–PSG–50HD.mov        [ 200 ]   → 200.SWS          ×
EPL–ARS–50HD.mov        [ 201 ]   → 201.SWS          ×
TNTS201  (30 frames)    [ 202 ]   → 202.SWS          ×
```

This replaced the old **Start number** and **Start slot** boxes in v1.8.0. Before that, batch numbering was invisible: you set a starting number, pressed Convert, and found out what each file had been called by reading the log afterwards. Now you can see it, and change it, before anything is written.

The list appears for **every** output, not only numbered ones. K-Frame TGA and TGA Sequence name their output from the source filename, so those rows are shown greyed and read-only — you cannot change them, but you can see what you are getting.

4.4 Numbering

Type a number into each row, or tick **Number sequentially** and let MacHuna fill them in.

Number sequentially, ticked	A number appears in the first row and the rest follow from it. Change the first row and everything below re-numbers. The other rows are locked until the first has a value, so it is clear where to type
Unticked	Every row is yours to fill in. This is the default: filling a scattered set of empty slots on the desk is at least as common as numbering a batch straight through

The setting is remembered between sessions. The **numbers are not** — every batch starts from 1 unless you say otherwise.

> **Why numbers do not carry over.** Until v1.8.0 the Start number advanced automatically after each batch and persisted between sessions. That only made sense while the numbers were invisible: wipes rarely stay in the destination folder, so carrying on from where the last batch finished was usually noise. Now that every number is on screen before you convert, there is nothing to remember on your behalf.

Sony MVS TGA works the same way but takes a **4-character clip name** per row instead of a number, and has no sequential option — names cannot sensibly be counted upwards.

K-Frame EIF takes a slot number per row and writes `0001.eif`, `0002.eif` and so on.

Checks before anything is written

Three checks run when you press Convert, and any of them stops the batch:

1. **Nothing may be blank**, or outside 1–9999
2. **No two rows may share a number or name**
3. **Nothing may collide with the destination folder** — an existing `12.SWS`, whether a file or a split-file folder, an existing `0012.eif`, or a Sony clip-name folder

There is no overwrite option. Every offending row is marked in red with the reason (`duplicate number`, `already in destination`), the list scrolls to the first one and puts the cursor in it, and correcting a duplicate clears its partner's mark too.

4.5 After Converting

The list stays on screen. Converted rows lock and show ✓ **done**; anything that failed shows **✗** and the reason.

```
BAY–PSG–50HD.mov        [ 200 ]   ✓ done              ×
EPL–ARS–50HD.mov        [ 201 ]   ✓ done              ×
TNTS201  (30 frames)    [ 202 ]   × no video stream   ×
```

Convert only ever runs rows without a status, so nothing is converted twice. You can add more files to a finished batch and press Convert again: only the new ones run, and if you are numbering sequentially they carry on from the highest number in the list rather than starting again.

- × removes a single row. With sequential numbering on, the rows below close the gap; with your own numbers typed, they keep them. A converted row never renumbers — the file it wrote already exists
- **Clear All**, below the list, empties everything and starts again

> **Nothing is converted twice.** Before v1.8.0, MacHuna emptied the list after every conversion to guarantee this, because items could only ever be *added* to a selection. Locking converted rows gives the same guarantee while keeping the list as a record of what was done.

4.6 Cancel

Click **Cancel** during a conversion to stop after the current file. The conversion log is not written if cancelled mid-batch.

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5. Kahuna SWS Output

When outputting to Kahuna SWS, the following options are available:

Option	Description
Standard	Target video standard. 1080p/50 is most common for UK/European broadcast. Every SWS is written at 1920×1080: sources of other sizes are scaled (v1.12.1), interlaced material one field at a time.
Split >4GB	On by default. Files over 4GB are split into 2GB FAT32-safe chunks. See Section 12.
Ignore alpha	No key plane is written. Output is fill-only, matching K-Watch no-alpha behaviour. Use for fill-only content or when the Kahuna output does not use a downstream keyer.

Option	Description
Auto play	Sets the Auto Play flag in the SWS header. The Kahuna begins playback when the clip is loaded.
Loop play	Sets the Loop Play flag in the SWS header. The Kahuna loops the clip continuously.
TGA source interlaced	See Section 8.2.
Include audio	Embeds audio from the source into the .SWS file. Shown only when audio is detected in the source. See Section 9.

> **SWS → SWS standards conversion:** when the source is itself an SWS file, the output header's clip name follows the source SWS's name, and the output key state follows the source (a keyless source produces keyless output). **Audio is carried through** (from v1.12.0; earlier versions dropped it). The duration does not change when the standard does, so the sound comes across unchanged at the new frame rate. Untick **Include audio** to leave it out. **The picture stays 10-bit** (v1.12.1): a straight copy, a frame-rate change and a weave copy the source's own picture exactly, and deinterlacing works on the 10-bit picture (earlier versions passed through 8-bit TGA files, which lost precision). An interlaced SWS going to the same interlaced standard passes straight through (before v1.12.1 its fields were woven a second time, halving the clip); to a different interlaced rate it is refused.

5.1 Progressive to Interlaced (P→I)

When converting a progressive source to an interlaced standard, MacHuna field-weaves pairs of progressive frames into genuine interlaced frames. The frame count halves — a 50fps progressive source becomes 25fps interlaced output. This is the correct behaviour for the Kahuna.

Example: 100 frames of 1080p/50 source → 50 frames of 1080i/50 output.

The source must run at the interlaced field rate (double the frame rate). Weaving pairs of frames only keeps the right duration when the source is a genuine field-rate stream:

Progressive source	Interlaced target	Result
50p	1080i/50	✅ correct — weaves to 25 interlaced frames
59.94p	1080i/59.94	✅ correct
60p	1080i/60	✅ correct
25p	1080i/50	⛔ blocked — would play at 2× speed
29.97p	1080i/59.94	⛔ blocked
30p	1080i/60	⛔ blocked

If you give MacHuna a **same-rate** progressive source (e.g. a 25p file destined for 1080i/50) or any other incompatible rate, it will **stop with a clear error and write no file**, rather than silently produce a clip that plays at the wrong speed. Convert the source to the correct field rate first (e.g. 25p → 50p), or choose a progressive output standard. *(This rate check applies to video-clip, SWS→SWS and clip→TGA conversions, which carry a known frame rate. A loose TGA image sequence has no frame rate to check and is still assumed to be a double-rate field stream.)*

5.2 Interlaced to Progressive (I→P)

When converting an interlaced source to a progressive standard, MacHuna bob-deinterlaces to produce the correct number of progressive frames. Playback speed on the Kahuna is preserved.

Example: 50 frames of 1080i/50 source → 100 frames of 1080p/50 output.

Cross-rate conversions (corrected in v1.6.12). Deinterlacing doubles the frame count, which lands on the target rate exactly only when the progressive standard is double the interlaced source's frame rate — 1080i/50 → 1080p/50, or 1080i/59.94 → 1080p/59.94. Any other pairing needs a rate change on top, and before v1.6.12 MacHuna did not apply one, so the clip played fast or slow:

Source	Target	Before v1.6.12	Now
1080i/50	1080p/50	correct	correct
1080i/50	1080p/60	ran fast	correct
1080i/59.94	1080p/50	ran fast	correct
1080i/59.94	1080p/25	ran 20% slow	correct

Progressive to progressive at a different rate (v1.12.1). A 25p clip going to 1080p/50, a 50p clip to 1080p/25, or 59.94p going to a 50fps EIF, is converted to the new rate - frames repeated or dropped - so it keeps its duration and its sound still fits. Every output frame shows the source frame that is on screen at that moment, so the first frame always comes first, and 29.97 and 59.94 are worked out exactly (30000/1001 and 60000/1001), so long American clips do not drift. Every route picks the same frames. Earlier versions kept every frame and stamped the new rate, so the clip played fast or slow. **Interlaced to a different interlaced rate** (1080i/50 material to 1080i/59.94) is refused, as progressive to interlaced already is: changing a field rate needs a proper standards converter.

59.94 and 60 count as one family. Material at one can go to a standard at the other - 59.94p woven into 1080i/60, or a 1080i/59.94 SWS or 29.97i clip going to 1080i/60, and the same the other way. The frames go through untouched, so the clip plays 0.1% fast (or slow): a 10-second clip finishes about a hundredth of a second early, which nobody can see. The log says so each time.

This applies to video clips, SWS→SWS, and TGA outputs. **A loose TGA image sequence is the exception:** it carries no frame rate anywhere, so MacHuna assumes the source standard matches the output family you pick (an interlaced pile aimed at 1080p/50 is treated as 1080i/50 material). That assumption is right for normal use. If you are converting a TGA sequence across families — interlaced 50Hz frames aimed at a 60Hz progressive standard — convert via a video clip or SWS instead, so MacHuna has a real frame rate to work from.

6. K-Frame EIF — Native K-Frame Format

> **Confirmed on a live K-Frame (7 October 2026).** MacHuna's EIF files were loaded on a K-Frame at 1080p50 and at 1080i 25Hz, with and without audio, and played correctly: right speed, right colour, key wiping in and out. A MacHuna EIF looked identical to the desk's own import of the same clip. Several fixes came out of that session and are in v1.12.0: most importantly, **25fps (50i) EIF files made by earlier versions play too fast on a K-Frame**. Convert them again with v1.12.0.

MacHuna can read and write Grass Valley K-Frame `.EIF` files — the native clip format used by the K-Frame Image Store and ClipStore. The Image Store lives in the **frame**, not the control panel, so `.EIF` applies equally whether your surface is a **Kayenne, a Karrera or a GV Korona**. The format was fully reverse-engineered from real Kayenne-produced files.

6.1 What is EIF?

`.EIF` is the K-Frame's native clip container. Each file holds a complete clip: fill and key combined in a single proprietary pixel format, always 1920×1080, at 25fps or 50fps. **Interlaced (50i) material is stored as woven 25fps frames** - both fields in each frame - which is how the K-Frame itself stores it. Files are named by slot number: `0001.eif`, `0002.eif`, and so on. The desk takes the slot from the file name and shows the clip name stored inside the file.

Audio lives in a separate file. A K-Frame keeps a clip's sound in a companion `.eaf` file with the same number: `0001.eif` and `0001.eaf`. The desk pairs them by name, so **always copy both**. The `.eaf` holds four channels of 24-bit audio at 48kHz, left on channel 1 and right on channel 2.

6.2 Converting TO EIF

Any of the following inputs can be converted to K-Frame EIF:

Input	Notes
MOV, MP4, MXF, MKV, AVI	Alpha channel (fill + key) preserved if present
TGA sequence	Progressive or interlaced source — see TGA source interlaced option
Kahuna SWS	Lossless direct YCbCr repack, no RGB conversion. The SWS audio comes across into the `.eaf`

Output options:

Option	Description
Slot (per row)	Each row in the item list carries its own slot number, written as `0001.eif`, `0002.eif` and so on. Tick **Number sequentially** to fill them in from the first row. See Section 4.4.

Option	Description
TGA source interlaced	Shown when source is a TGA sequence. Tick when the TGA frames are interlaced. MacHuna keeps the woven frames exactly as they are and writes a 25fps EIF, which is how a K-Frame stores 50i. (Before v1.12.0 this deinterlaced to 50fps progressive, which did not match the desk.)
Include audio	On by default when the source has sound. Writes the companion `.eaf` : the source's first two channels become left and right, and a mono source goes to both. 24-bit audio keeps all 24 bits. Untick it for a silent clip.

EIF output is always 1920×1080. Sources of other sizes are scaled (interlaced TGA frames one field at a time, so the fields never blend). Frame rate is rounded to the nearest EIF-supported rate (25fps or 50fps), and the video is resampled to that rate so the clip keeps its original duration and plays at the correct speed. For example a 60fps source becomes a 50fps EIF clip of the same length.

The picture and its audio are one output. When MacHuna writes `0005.eif` it either writes a fresh `0005.eaf` or, if this clip has no sound, removes an old `0005.eaf` left from an earlier conversion, and says so in the log. Otherwise a desk could play last week's sound with today's picture. **If you cancel, or a conversion fails**, nothing is left half-written and any earlier clip of the same name is left exactly as it was (v1.12.1): the `.eif` and `.eaf` are written under temporary names and only put in place, together, once both are complete. (In v1.12.0 a cancel deleted the half-written file - and with it the earlier clip.)

6.3 Converting FROM EIF

EIF files can be converted to the following outputs:

Output	Notes
Kahuna SWS	Lossless direct YCbCr repack, to the **Standard** you choose (v1.12.1): a 25fps EIF can be 1080p/25 or 1080i/50 (choose 1080i/50 for 50i material, which is what a 25fps EIF usually is); a 50fps EIF can be 1080p/50, or 1080i/50 by weaving pairs of frames; other progressive standards are converted to the new rate; 59.94Hz and 60Hz interlaced are refused. The key is always kept: an EIF always carries one, and a key that is opaque throughout may be meant, to keep a clip full frame for its whole length. The clip's `.eaf` audio is carried into the SWS on the Kahuna's channels (left 1, right 3); untick **Include audio** to leave it out.
K-Frame TGA	Full-resolution 1920×1080 32-bit RGBA TGA sequence. Frames numbered `0001.tga` onwards.
Sony TGA	32-bit RGBA TGA sequence with 4-character clip name prefix.

For TGA outputs, see Section 7.4: MacHuna works out from the EIF whether anything needs weaving. A 25fps EIF already holds woven 50i and goes through frame for frame. The same field order and clip name options as regular extraction apply - see Section 7.

6.4 EIF in the Video Player

The built-in Video Player opens `.EIF` files directly. Frame rate is detected automatically from the file header, with no prompt. Fill, key and composite panels are all populated, and **the clip's `.eaf` audio plays** (from v1.12.0; earlier versions played EIF clips silent). See Section 10.

6.5 Loading ProRes straight into a K-Frame

Something found during the K-Frame testing that is worth knowing even if you never use MacHuna for it. **A K-Frame's own MOV import treats every ProRes file as SD colour (BT.601)**, whatever the file is labelled, so saturated colours shift: reds come out darker, blues move. Tested with ProRes labelled RGB, unlabelled, and properly labelled HD; the desk's result was identical each time, so no export setting avoids it. QuickTime Animation (RGB) files import correctly.

Converting through MacHuna keeps the colour accurate. MacHuna reads a labelled file by its label, and an unlabelled HD file (720 lines or more) as HD colour, BT.709, which is the industry convention. For ProRes deliveries, convert to EIF in MacHuna rather than importing the MOV on the desk.

The log says when it has made that choice (v1.13.1): "Source has no colour label, treated as HD colour (BT.709)". Grass Valley's own K-Watch makes the opposite choice for an unlabelled file, so the same unlabelled clip converted by each can look different on the desk; a labelled file looks the same from both.

Two things to know. **Before v1.12.1, MacHuna read an unlabelled HD file as SD colour, just as the desk does**, so a conversion of such a file made with an earlier version has the same shift; convert it again. And if a file really is SD colour at HD size but carries no label, MacHuna cannot tell: label it when you export it.

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7. Extraction Outputs — K-Frame and Sony MVS

> **K-Frame TGA is confirmed on a live K-Frame (7 October 2026)**, from video clips, `.EIF` and `.SWS`. On the desk, import it through **Image Store > Library > From Disk/Folders** and **select "Sequence"**: without that, the desk lists every frame as a separate still. Compressed and uncompressed TGAs are both accepted.
>
> *QuickTime MOV is listed separately below. It is not a desk format.*

7.1 Output Targets

Output	Format	Destination desk
K-Frame TGA	32-bit RGBA TGA sequence. Frames numbered `0001.tga` onwards. One subfolder per SWS.	Grass Valley K-Frame Image Store

Output	Format	Destination desk
Sony TGA	32-bit RGBA TGA sequence. Frames numbered `XXXX0000.tga` (4-character clip name + frame number). One subfolder per SWS.	Sony MVS Image Store
QuickTime MOV	ProRes 4444 `.mov`, key carried as a real alpha channel, audio included where the source has it. Named after the source file.	None — an ordinary video file for an edit suite

7.2 Workflow

1. Open a folder of `.SWS` files
2. Select the output target from the **Output** dropdown
3. Set any options shown (Standard, Clip name, Field order, Include audio)
4. Click **Convert**

7.3 QuickTime MOV — getting material back out

Added in v1.10.0. Converts a Kahuna `.SWS`, a K-Frame `.EIF` or a TGA sequence into a **ProRes 4444 QuickTime**: the fill as picture, the key as a real alpha channel, and sound where the source has any. The file is named after its source, so `51.SWS` becomes `51.mov`.

This is the way to get a graphic out of a desk format and into an edit suite, grade or archive. It is **not** a desk format, so unlike K-Frame TGA or Sony TGA there is no hardware warning on it and nothing awaiting confirmation — if it opens in QuickTime or your NLE, it is right.

Where the audio comes from:

Source	Audio
`.SWS`	Read from the file itself, which carries audio inline
`.EIF`	Read from the clip's companion `.eaf` file, if it has one. A K-Frame keeps clip audio in a separate file of the same name
TGA sequence	None — a sequence of images has no audio to carry

Untick **Include audio** to produce a silent MOV from a source that has sound.

Naming the file. Each item in the list has a name field. Leave it blank and the MOV is named after its source, which is usually what you want: `1.SWS` becomes `1.mov`, a sequence called `MYSEQ` becomes `MYSEQ.mov`. Type a name and it uses that instead, so a TGA render can come out as `Barcelona Wipe.mov` rather than inheriting whatever the render was called.

Names may contain spaces, letters, digits and ordinary punctuation, up to 64 characters. Anything a filesystem would refuse is blocked as you type. Two items cannot be given the same name, and a name that already exists in the destination is refused rather than overwritten — the same protection the numbered outputs have. Typing `.mov` on the end is harmless; it will not be doubled.

Frame rate. An `.SWS` and an `.EIF` both declare their rate in the header, so MacHuna uses it and no control appears. A TGA sequence declares nothing, so for TGA input a **Frame rate** control appears: 23.976, 24, 25, 29.97, 30, 50, 59.94 or 60. Set it to the rate the sequence was made at.

> **Fixed in v1.10.2.** This used to be the **Standard** dropdown, which offers only the seven verified Kahuna broadcast standards. Its only 30fps entries are *interlaced*, so a 30fps sequence had nowhere correct to go — picking a progressive standard played it at the wrong speed, and picking an interlaced one wove fields into a QuickTime. A MOV is an ordinary video file and now gets ordinary video rates. **The "TGA source interlaced" tickbox would not have helped either** — ticking it left the clip running at double speed and put progressive frames through a deinterlacer.

If your TGA frames are themselves interlaced, tick **TGA source interlaced** and choose the field order. MacHuna deinterlaces them into full progressive frames — one frame per field, so a 25fps interlaced sequence becomes a 50fps progressive MOV of the same length, with smooth motion and no combing. Leave it unticked for ordinary progressive renders, which is the normal case.

> A QuickTime is never *woven* into interlace — ProRes carries no field-order flag, so there would be nothing to set and nothing downstream to read it. But frames that already contain fields still have to be *deinterlaced*, or the comb stays in the picture. Those are two different things, and v1.10.2 briefly dropped both. Restored in v1.10.3 after David spotted it.

Stills are not accepted, in line with the rest of MacHuna: a single frame is not a clip. Stills convert to Kahuna SWS only.

> **Interlaced sources.** ProRes carries no field-order flag, so an interlaced source produces a MOV with correctly decoded interlaced frames that a downstream NLE may not automatically identify as interlaced. Set the field order in your NLE if it matters.

7.4 Standard (TGA outputs)

A TGA sequence has no frame rate of its own: the desk imports the frames at whatever format it is set to. So MacHuna works out from the **source** whether anything needs doing (v1.12.0):

Source	With an interlaced standard
Already interlaced (a 50i clip, a 1080i SWS, a 25fps EIF)	Frames go through exactly as they are
Progressive at 25fps	Frames go through exactly as they are
Progressive at 50fps or more	Pairs of frames are woven into interlaced frames, for a 50i desk

With a progressive standard, frames always go through as they are. MacHuna logs what it did for each file.

Every K-Frame TGA and Sony TGA frame is **1920×1080, 32-bit RGBA**, whatever the source (v1.12.1). A video clip of another size is scaled (an interlaced one a field at a time, so the fields never blend), and a clip with no alpha gets a fully opaque one. Before v1.12.1, frames from a video clip kept the clip's size and were 24-bit when it had no alpha.

> **Fixed in v1.12.0.** An interlaced video clip used to be woven a second time on an interlaced standard: half the frames, each mixing two moments in time. This affected K-Frame TGA and Sony TGA. A 25fps EIF used to be refused with an error.

7.5 Field Order (TGA outputs)

A **TFF / BFF** toggle appears for interlaced standards, and always for Sony TGA. **TFF (Top Field First) is the default** — it is the SMPTE standard for HD (including 1080i) and is correct for all known 1080i HD workflows (default since v1.6.3). If you see motion artefacts or comb effects on the desk after import, switch to BFF and reconvert.

> **Fixed in v1.6.12.** For Sony TGA output from a TGA sequence or a video clip, this toggle was displayed but had no effect — the conversion was hardcoded to TFF, so switching to BFF changed nothing in the output. It now works in both conversion directions, and the conversion log states which field order was applied, so you can confirm what was used. If you tested Sony TGA output before v1.6.12 and concluded BFF did not help, that test was not valid and is worth repeating.

7.6 Sony TGA — Clip Name

Enter a **clip name of exactly four letters or digits** (e.g. `WIPE`). Anything else is refused before anything is written (v1.12.1; a shorter name used to be padded with spaces in every frame name, and a longer one cut short without a word). All TGA frames in the batch share this name — on the Sony MVS, files with the same 4-character prefix are grouped into a single clip on import.

Each clip takes its own 4-character name from the item list (see Section 4.4), so each becomes its own output folder and several Sony clips convert together. The shared Clip name field that once forced one clip at a time is gone as of v1.8.0.

7.7 Output Structure

Output	File naming
K-Frame TGA	Subfolder per source, named after its file name. Frames `0001.tga` … inside.
Sony TGA	Subfolder per SWS, named after the 4-character clip name. Frames `XXXX0000.tga` … inside.

8. TGA Sequences

TGA sequences are multi-frame clips stored as individually numbered still images. MacHuna handles them through the folder browser — you do not need to select individual frames.

MacHuna accepts any numbered TGA naming convention — K-Watch names, After Effects exports, third-party renders, or anything else:

```
TNTS201_30_0001.tga   (K-Watch naming)
FEDX0000.tga          (letters + digits, no separator)
shot_0001.tga         (underscore separator)
render.0001.tga       (dot separator)
```

The only requirement is that frames are numbered sequentially.

8.1 In the Folder Browser

When you open a folder containing a TGA sequence, MacHuna collapses the entire sequence to a single entry:

```
TNTS201  (30 frames)
```

Select this entry and convert as normal. MacHuna handles the frame ordering automatically.

8.2 TGA Source Already Interlaced

If you are re-wrapping TGA frames that were previously extracted from an interlaced SWS (e.g. via MacHuna's extraction outputs), tick **TGA source interlaced**. This tells MacHuna to pass frames through directly without applying field-weaving. Without this, MacHuna would incorrectly treat the already-interlaced frames as progressive and weave them again.

When **TGA source interlaced** is ticked and you select a **progressive** Kahuna SWS standard (e.g. 1080p/50), MacHuna deinterlaces the frames using yadif (`send_field` , TFF per SMPTE 274M) rather than duplicating them: each interlaced frame is separated into two progressive fields, doubling the frame count with correct, smooth motion. Any alpha/key channel is deinterlaced with the identical filter so fill and key stay aligned.

For K-Frame EIF output it is different: the woven frames are kept exactly as they are, at 25fps, because that is how a K-Frame stores 50i (see Section 6.2).

8.3 TGA Sequence Output

When the input is a TGA sequence or a video clip, **TGA Sequence** appears as an output option. This re-writes the frames as a new numbered TGA sequence in a subfolder, applying an interlaced⍰progressive conversion according to the selected **Standard** — useful for converting a sequence between field standards without going through SWS. Progressive→interlaced uses field-weaving (`tinterlace`); interlaced→progressive uses yadif deinterlacing. Both follow the **Field order** toggle where it is shown, and default to TFF otherwise. The alpha/key channel is preserved throughout.

8.3 Alpha Channel

- If TGA files have an alpha channel, the key plane is extracted automatically
- If no alpha is present, no key plane is written (from v1.11.0, confirmed on a live Kahuna; earlier versions wrote a flat key plane that keyed out to nothing)
- If Ignore alpha is ticked, no key plane is written regardless

8.4 Audio

Audio is not supported in TGA sequence conversions.

---

9. Audio

MacHuna extracts audio from source files and embeds it in the `.SWS` file in K-Watch native format.

9.1 Audio Format

Encoding	16-bit signed PCM, big-endian, one 4-byte slot per channel
Sample rate	48,000 Hz
Channels	8 slots per sample (32 bytes)
Channel mapping	Ch1 = Left, Ch2 = Right, all others silent
Length	The sound's own length, whatever the picture's (v1.13.0). The header says exactly how many samples there are, and a Kahuna plays that many

Confirmed on a live Kahuna (9 October 2026), by ear. This is the layout K-Watch writes. It was settled by comparing K-Watch conversions that play correctly on a Kahuna with the same sources, sample by sample. **Before v1.12.4, MacHuna wrote SWS audio in a different layout that a Kahuna played for only the first eighth of a second, at reduced quality: convert any SWS with sound made by an earlier version again.** The picture and key in those files were always fine. MacHuna still reads its older files correctly (in the player, and converting them to other formats).

The sound keeps its own length (v1.13.0), as Grass Valley's Kahuna Manager writes it: a clip whose sound runs longer than its picture keeps all of it, and on the desk the picture holds its last frame while the sound plays on. Before, the sound was cut or padded to the picture. A K-Frame `.eaf` is still fitted to the picture, because that is what the K-Frame does. (Adding a separate sound file to a clip or a TGA sequence, and saving a clip's sound as a WAV, are MacHuna 2 features.)

Mono goes to both channels (from v1.12.0). Earlier versions put a mono source on the left only. Stereo and multichannel sources are unchanged: the first two channels are left and right. **Dual mono** - two separate mono tracks, common in broadcast ProRes - is track 1 left, track 2 right (v1.12.1; before, track 2 was dropped).

9.3 K-Frame audio (`.eaf`)

File	Companion `.eaf` beside the `.eif`, same number
Encoding	24-bit samples in 32-bit little-endian words
Sample rate	48,000 Hz
Channels	4. Ch1 = Left, Ch2 = Right, Ch3-4 silent
Samples per frame	1920 at 25fps, 960 at 50fps
Length	Padded with silence, or trimmed, to exactly the clip length

Confirmed on a live K-Frame (7 October 2026): the desk accepted a MacHuna-written `.eaf` and gave it back unchanged, and MacHuna's `.eaf` for a test clip was identical to the one the desk made itself. **The `.eaf` keeps all 24 bits; the K-Frame's own MOV import keeps only 16.**

Every route keeps the audio the target expects. MOV to EIF and SWS to EIF write the `.eaf`; EIF to SWS and SWS to SWS carry the audio into the SWS; EIF to QuickTime MOV writes 24-bit audio, so nothing is lost. A 24-bit sound going into a 16-bit SWS keeps its top 16 bits, as the K-Frame does.

9.2 Notes

- Audio is appended after the key plane in the `.SWS` file
- If the source has no audio and Include audio is ticked, the option is simply ignored — no error
- Audio detection in third-party `.SWS` files uses the audio offset and format flag fields (0x1E8 and 0x1EC). The audio frame size field at 0x1C2 is unreliable across workflows and is not used for detection

---

10. Video Player

The built-in Video Player lets you check a file without needing a Kahuna or K-Frame desk. Click the **Video Player** button to open it.

10.1 Accepted Inputs

Input	Fill / Key / Composite	Audio
`.SWS`	Yes	Yes (if present in file)
`.SWS` **split files** (a folder of `01_OF_03._XX` parts)	Yes	Yes (v1.13.1; earlier MacHuna splits were written without sound)
`.EIF` (K-Frame native)	Yes - fill, key, and composite all populated	Yes, from the companion `.eaf` (v1.12.0)
`.TGA` sequence (pick any frame)	Yes (if TGAs have alpha)	No

Input	Fill / Key / Composite	Audio
`.MOV`, `.MP4`, `.MXF`, `.MKV`, `.AVI`	Yes (alpha preserved for ProRes 4444 etc.)	Yes

Click **Open…** and select a file. For TGA sequences, pick any frame from the sequence — MacHuna loads the whole sequence.

> **If a clip looks split down the middle (v1.9.1).** Some K-Watch files place the video after a larger header than usual, and MacHuna used to assume a fixed position — reading every frame half a line early, which made the picture appear cut in half with the sides swapped. It now reads the position from the file. If you have converted any file that displayed this way, convert it again: the same fault affected K-Frame EIF, K-Frame TGA, Sony TGA and TGA sequence output, not just the player.

Split files (v1.9.0). A clip over 4GB is stored as a *folder* named `<n>.SWS` containing `01_OF_03._XX`, `02_OF_03._XX` and so on. **Select the folder if your file dialog lets you, or open it and pick any one of the parts** — either loads the entire clip, not just the part you clicked. The info strip confirms it with `Split: 3 parts`. Before v1.9.0 these could not be opened at all: the dialog treats the folder as somewhere to navigate into, so you reached a single chunk, and only the first chunk carries a header. When opening a TGA sequence, you will be prompted for the frame rate (25fps default). For EIF files, the frame rate is detected automatically from the header.

10.2 Display Layout

Panel	Content
Fill	The video content plane
Key	The alpha/key plane (greyscale)
Composite	Fill composited over a chequerboard using the key as alpha — shows transparency
Audio	VU meters for left and right channels

10.3 Transport Controls

Control	Action
Cue	Jump to the first frame
Play	Begin playback at the clip's native frame rate
Pause	Pause at the current frame
Stop	Stop and return to the first frame

10.4 Playback Stutter

Playback can stutter for the first few plays of a clip, then settles. This is a limitation of the player, not a fault in your file — if a clip stutters on the first pass and plays cleanly on the second, the clip is fine.

The player exists as a confidence check: to see that a clip is what you expected before it goes to a desk. It is not a broadcast playout tool, and smooth first-pass playback would need it rebuilt around a proper media view. That is a job for the native rewrite rather than something to patch here.

10.5 Notes

- Multiple player windows can be open simultaneously
- Split `.SWS` files (>4GB, multi-chunk) play as one clip (v1.9.0), sound included (v1.13.1)
- Audio playback uses the sounddevice library. If not installed, the player opens without audio but VU meters are still drawn

---

11. Updates and Getting in Touch

Everything in this section lives in the **Help** menu, which is new in v1.7.0.

11.1 The Help Menu

Item	What it does
MacHuna User Manual	Opens this document. It is bundled inside the app, so it works with no internet and always matches the version you are running
Check for Updates…	Checks now, and always tells you the result
Report a Problem…	Opens an email with the details already filled in
Suggest a Feature…	The same, for things MacHuna does not do yet

11.2 Update Notifications

A few seconds after MacHuna opens, it reads a small file at `dnsvision.tv/machuna/version.json` in the background to see whether a newer version exists. This never delays the window opening.

If a newer version is available, a strip appears across the top of the MacHuna window with the version number, a line about what changed, a **Download** button and a **Dismiss** button. Download opens the download page in your browser; MacHuna never installs anything itself.

If you are up to date, offline, or the check fails for any reason, nothing at all appears. No error, no delay, no message. This is deliberate. MacHuna is used in OB trucks and galleries with no route to the internet, and an app that hangs on startup or nags about a network it cannot reach would be worse than useless. The request gives up after five seconds and is forgotten.

Dismiss hides the strip for that session. Choosing **Check for Updates…** from the menu shows it again, because asking is a request to be told.

> **What is actually sent: nothing about you.** MacHuna reads one small public file and nothing else. It sends no information about you, your machine, your files or your conversions, and there is no usage tracking of any kind. The request is an ordinary web fetch, the same as opening a page in a browser.

Check for Updates… does the same check but always reports back: a newer version, "You're on the latest version", or "Couldn't check for updates just now". Only the automatic check on launch is silent.

11.3 Hardware-Status Notices

From v1.7.1 the log noted, at the head of each batch, any output that had never been loaded on the desk it is for. **As of v1.12.0 there are none.** K-Frame EIF and K-Frame TGA were confirmed on a live K-Frame on 7 October 2026, Kahuna SWS has been confirmed for a long time, and the Sony MVS note has been withdrawn. The "has not been tested on hardware - proceed anyway?" questions before EIF output and video-to-TGA are gone too.

The full status of every output is also published at dnsvision.tv/machuna.

11.4 Reporting a Problem

Help ▸ Report a Problem… opens a panel showing exactly what is about to be shared, then hands it to your own mail app as a draft.

The draft arrives already carrying the things that otherwise take three emails to establish: the MacHuna version, your macOS version, your Mac model, and the standard, output format and source files you had selected. You write what happened in your own words in the spaces at the top.

Three buttons:

- **Open Email** — opens the draft in your mail app. **Nothing is sent until you send it.** You can edit or delete any part of it first, and on a truck with no signal it simply waits in your outbox
- **Copy Details** — copies the same text to the clipboard, for anyone whose Mac has no mail app set up
- **Show Log in Finder** — appears when a conversion log exists in your destination folder. That log usually answers the question on its own, so attaching it helps

Suggest a Feature… is the same panel with different prompts, for a format, a standard or a naming convention your facility needs.

Both go to **machuna@dnsvision.tv**.

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12. Large File Support (>4GB)

Files larger than 4GB are automatically split into 2GB chunks when **Split >4GB** is enabled (on by default). The split format exactly matches K-Watch output and has been confirmed working on a live Kahuna mainframe.

12.1 Split File Structure

```
201.SWS/
  01_OF_03._XX   (512-byte header + video data, exactly 2GB)
  02_OF_03._XX   (video data, exactly 2GB)
  03_OF_03._XX   (video data, remainder)
```

- The parts are one ordinary SWS file cut into 2GB pieces: header, fill, key, then sound
- Chunk 1 starts with the SWS header; the sound, if any, is at the end of the last part
- The header in chunk 1 carries the total frame count across all chunks

> **Convert keyed split clips again if they were made before v1.13.1.** Earlier versions wrote every split file with a header saying it had no key, and left its sound out. Confirmed on a Kahuna on 2026-10-12: such a clip loads with **no key at all**. A v1.13.1 split, of the same clip as K-Watch's own, played identically on the desk: key, sound and length. Clips under 4GB, and clips with no key, are not affected by the key fault.

> **IMPORTANT** When transferring to a Kahuna via USB, copy the entire `.SWS` folder (e.g. `201.SWS/`), not the individual chunk files inside it.

13. SWS Format Reference

This section is for support engineers and developers. It documents the SWS binary format as reverse-engineered from K-Watch reference files and verified against a live Grass Valley Kahuna mainframe.

13.1 File Layout

Range	Content
`0x000 – 0x1FF`	512-byte header (big-endian)
`0x200 – N`	Fill plane: v210 big-endian, `plane_size × frame_count` bytes
`N – M`	Key plane: v210 big-endian, same size as fill plane (absent if `play_count == 0`)
`M – EOF`	Audio data: 16-bit LE PCM, 16ch, 48kHz (absent if audio offset == 0)

13.2 Key Header Fields

Offset	Type	Description
`0x188`	uint32 BE	Video standard code (OR'd with playback flags — see 13.4)
`0x18C`	uint32 BE	Format variant code (unambiguous fps lookup — all nine values are unique)
`0x190`	uint32 BE	Width in pixels
`0x194`	uint32 BE	Height in pixels (fill plane)
`0x1A0`	uint32 BE	Plane size (bytes per frame, fill or key)
`0x1A4`	uint32 BE	Frame count
`0x1A8`	uint32 BE	Play count (= frame count; 0 if no key plane)
`0x1B4`	uint32 BE	`(plane_size × frame_count + 512) / 32` ; 0 if no key
`0x1C2`	uint16 BE	Audio frame size — unreliable, do not use for detection
`0x1CC`	uint32 BE	Total file size in bytes
`0x1E8`	uint32 BE	Audio data offset ÷ 32 (0 if no audio)
`0x1EC`	uint32 BE	Audio format flag: `0x03000000` if audio present

13.3 Audio Detection

Reliable method: `aud_offset (0x1E8) > 0` AND `aud_fmt (0x1EC) == 0x03000000`.

The four bytes at `0x1C0` are the **number of audio samples**, and a Kahuna plays that many and no more (confirmed 9 October 2026). Earlier versions of this manual called the last two of them an "audio frame size" with a fixed value of `0x1680`; that was one reference file's sample count, mistaken for a constant.

13.4 Playback Flags

Bits 2 (`0x04`) and 3 (`0x08`) of the low byte at `0x188` are OR'd into the video standard code:

Flag	Bit	Value
Auto Play	2	`0x04`
Loop Play	3	`0x08`

13.5 Interlaced Flag

Bit 15 (`0x8000`) of `0x188` is set for all interlaced standards. The standard code for all interlaced formats is `0xC923`.

14. Troubleshooting

Kahuna showing black key / no key

- If **Ignore alpha** is ticked, no key plane is written — correct behaviour
- If the source has no alpha channel, no key plane is written and the desk shows Fill only - correct behaviour
- If you are expecting a real key but getting none, check your source file has a valid alpha channel

Audio not playing on Kahuna

- Confirm **Include audio** is ticked
- Confirm the source file has an audio track (MacHuna hides Include audio if no audio is detected)
- Audio is resampled to 48kHz by ffmpeg if the source is at a different rate — this is automatic

File >4GB not loading on Kahuna

- Confirm **Split >4GB** is enabled
- Copy the entire `.SWS` folder to the USB drive, not the individual chunk files inside it
- The USB drive must be FAT32-formatted

Conversion fails with ffmpeg error

- Check the Log area for the ffmpeg error message
- Confirm the source file is not corrupted — try opening it in another application
- Confirm the Destination Folder path exists and is writable

Extraction output not loading on K-Frame or Sony MVS

- On a K-Frame, import TGA sequences through **Image Store > Library > From Disk/Folders** with **Sequence** selected; otherwise each frame appears as a separate still and the import buttons stay greyed out
- Check that the correct Standard is selected for the target desk's video format
- For Sony TGA: confirm the 4-character clip name matches your expected import workflow
- For interlaced output: if motion artefacts appear, try switching the Field Order toggle (TFF ? BFF; TFF is the default)

EIF file not loading or playing wrongly on K-Frame desk

- **Plays at double speed, possibly with a green bar across the top:** if the desk's format was changed recently, switch it to another format and back until a known-good clip plays correctly. On 7 October a desk changed to 1080i played every 25fps clip this way, its own included, until it was switched back and forth
- **A 25fps clip made before v1.12.0 plays too fast:** convert it again with v1.12.0
- **No sound:** copy the `.eaf` with the `.eif` - same number, same folder

- Confirm the file is named with a 4-digit slot number (e.g. `0001.eif`)
- Check the Log area for any conversion notes

TGA sequence not appearing as a single entry in the file list

- Confirm the TGA files are in the same folder with no other file types present
- MacHuna detects TGA sequences by parsing numbered filenames — files must be numbered sequentially. Any naming convention works (K-Watch, After Effects, custom renders, etc.) as long as the base name is consistent and frames are numbered.

File plays at double speed on Kahuna

This can happen if:

- A progressive source was converted to a Kahuna SWS without using the progressive standard, and there is a mismatch between the frame count and the interlaced header
- A TGA sequence from an interlaced source was converted without ticking **TGA source interlaced** — MacHuna field-weaved the already-interlaced frames, halving the frame count again

Check the conversion log for any warnings about P→I or interlaced detection.

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15. Known Limitations

- **Apple Silicon only** — Intel Mac builds are not supported
- **TGA sequence audio** is not supported
- **HLG Rec.2020** colour space is not implemented (requires a reference HLG .SWS file to reverse-engineer the header values)
- **An EIF does not record whether it is interlaced.** When converting a 25fps EIF to SWS, choose 1080i/50 for 50i material and 1080p/25 for progressive

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16. Authors

David Steer / DNS Vision Limited & Claude (Anthropic)

MacHuna was built collaboratively using AI-assisted development. The SWS format was reverse-engineered from K-Watch reference files and verified against a live Grass Valley Kahuna mainframe. The K-Frame EIF format was reverse-engineered from real Kayenne-produced clips and confirmed on a live K-Frame in October 2026, where the `.eaf` audio format was also settled.